

# Armaan Mittal

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## EDUCATION

### Thapar Institute of Engineering and Technology, Patiala

August 2022 – May 2026

- BE in Electronics and Computer Engineering with **minor in Conversational AI**.
- **Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems, Computer Networks, Machine Learning, Generative AI, Natural Language Processing.

## TECHNICAL SKILLS

**Languages:** C++, Python, Java, Go, SQL, Bash

**Systems & Agentic Tooling:** Computer Architecture, Memory & Concurrency Models, OS Internals, Data Structures & Algorithms, Claude Code, Model Context Protocol (MCP), Multi-Agent Orchestration, Tool-Calling

**Backend/Infra:** Relational Databases (MySQL/PostgreSQL), NoSQL (MongoDB), Distributed Systems, Low-Latency Systems, Linux, Git, CI/CD, Docker, Kubernetes

## EXPERIENCE

### Software Engineer (Implementation), Engagely.ai

June 2026 – Present

- Designed and built a **multi-agent orchestration system** (role-specialized agents coordinated by a central controller with **tool-calling**) on a **multi-tenant SaaS** platform (**Python, FastAPI, Celery, Redis**), using **RAG** for grounded responses.
- Deployed the platform for enterprise clients including **Cipla** and **ART**; debugged & resolved production issues across distributed **Celery/RabbitMQ** workers and **Azure OpenAI** integrations via **root-cause analysis**.
- Built **load testing** (Socket.IO/Engine.IO, **1000 virtual users**), implemented **SSO/OIDC** auth, and integrated **TTS/STT** (XTTS-v2) into production pipelines.

### Research Intern, Samsung R&D Institute Bangalore

Feb 2025 – Nov 2025

- Built low-latency NLP modules (**gibberish detection**, PII masking) for **Bixby** across 12 languages, achieving **93.4%** multilabel accuracy; built an **MCP server and client** to expose the tooling.
- **Deployed and monitored** low-latency LLM modules in production; reduced end-to-end response latency by **15%** via **quantization** and cache-aware preprocessing, with integration tests and monitoring dashboards.
- Productionized with **CI pipelines, Docker, Kubernetes**, collaborated with platform teams to tune limits and alerts.

### Student Researcher and ELC Intern, TIET Patiala

July 2025

- Collaborated on a research project aimed at **speech reconstruction** using brain EEG signals (66 channels); authored reproducible training scripts implementing Conformer, MAE, and Siamese components.
- **Reduced inference 74s → 2s** via pruning and streaming chunking for multimodal fusion.

## PROJECTS

### LatentBook | Low-Latency Order-Book & Market-Data Simulator

2026

- Built a **C++** limit order-book engine (price-time priority matching, lock-free ring buffers) simulating live market-data feeds, with a focus on **cache-aware memory layout** and predictable tail latency.
- Used **Claude Code** and custom **MCP tools** to accelerate development, building agent-driven workflows for benchmark generation, latency-regression detection, and test scaffolding across the codebase.
- Profiled and optimized p50/p99 latency under concurrent order flow; validated correctness with property-based and stress tests.
- **Technologies:** C++, Multithreading/Concurrency, Google Benchmark, Claude Code, MCP.

### EduSched | Advanced Timetable Management System for College

June 2025 – July 2025

- Built a **user-facing** internal scheduling platform (React/Node/MySQL) with **modular auth/scheduler/notifier services**; **cut reschedule overhead 70%** and improved notification turnaround 95%.
- Automated **alternative time/room suggestions** via precomputed feasible-slot indexing; delivered real-time notifications over WebSockets.
- **Technologies:** React, Node.js (Express), MySQL, WebSockets, Docker, Kubernetes.

## ACHIEVEMENTS

- **Accepted research paper (ASCML 2026):** “Scalable Multilingual Content Validation: A Dataset-Free Approach to Gibberish Detection”.
- **Paper under review, *Expert Systems With Applications* (Elsevier):** “PRISM-Home: Policy-Routed, Interpretable, State-Grounded Messaging”.
- Selected for **Amazon Summer School 2024** (national AI/ML program) and **National Finalist**, Meta Pragati Hackathon.
- **Head, Competitive Programming Contest** (Saturnalia Fest) organized and led national-level programming competition with 300 participants.
- Mentored 3K+ students in Data Structures & Algorithms via recorded modules and live sessions.
- Ranked **362 and 535** out of 20,000+ on CodeChef [[armaanmittal](#)].